

Assessment CompTia N+

Section A- Concept Application

Scenario 1

- **CPU:** multi-core (6+ cores) with decent clock speed — compiling code and running a browser with many tabs plus video encoding for conferencing are all CPU-hungry simultaneously. Check socket compatibility with the motherboard.
- **RAM:** 16GB minimum (32GB is safer) — IDEs, containers/ VMs, dozens of browser tabs, and conferencing apps each hold significant memory; also confirm RAM speed/type (DDR4 vs DDR5) matches what the motherboard supports.
- **Storage:** NVMe SSD, not HDD — faster boot, faster IDE indexing/compilation, faster app switching. Capacity should account for dev tools, VMs, and project files.
- **Compatibility check:** CPU socket ↔ motherboard socket, RAM type/speed supported by the motherboard's QVL, and enough PCIe lanes/M.2 slots for the SSD.

Scenario 2

- **CPU:** real-time scanning hooks every file read/write and process launch, adding constant background CPU overhead; scheduled full scans spike CPU usage further.
- **RAM:** the AV engine itself consumes memory, and it holds virus definitions/heuristics data in memory while running.
- **Storage:** every file access is intercepted and scanned before being allowed to complete, adding I/O latency — especially painful on HDDs, but noticeable even on SSDs during a full scan.

Scenario 3

1. Check Device Manager for the device — look for a yellow warning icon or "unknown device."
2. Note the exact error code / driver status.
3. Try a different USB port (rules out a dead port before blaming drivers).
4. Update or reinstall the device driver (uninstall in Device Manager, then reconnect to force Windows to re-detect).
5. Check OS/driver version compatibility — an older OS may lack native drivers for newer peripherals, or a driver may need a vendor update for that specific OS build.
6. Check Windows Update for optional driver updates.
7. Test the same device on a known-good known-working port/machine to isolate device vs. host.
8. If still failing, check BIOS settings (USB legacy support, port power settings) and update the motherboard's chipset drivers.

Scenario 4

Surge protector: absorbs and diverts sudden voltage spikes (e.g., from lightning or grid switching) so they don't reach and damage sensitive components like the PSU or motherboard.

UPS: provides continuous power from a battery during outages/brownouts, giving the system time to save work and shut down gracefully instead of losing power abruptly — this prevents data corruption and file-system damage. A line-interactive or online UPS also smooths out voltage fluctuations (sag/surge) in real time, not just during a full outage.

Scenario 5

DHCP automatically assigns each device an IP address, subnet mask, default gateway, and DNS server address when it joins the network.

DNS translates hostnames (e.g., appserver.nexus.local) into IP addresses so users don't need to memorize numbers.

Since devices **can reach systems by IP but not by name**, and **sometimes fail to get an IP at all**, DHCP is the more likely root cause: if DHCP is intermittently failing (scope exhausted, DHCP

server down/misconfigured, rogue DHCP server, short lease conflicts), some devices get no address, and others may get an address but not the correct DNS server option — which would also explain the name-resolution failure. So: **DHCP misconfiguration is the primary suspect**, with a secondary check needed on the DNS server address being handed out.

Scenario 6

1. Check the firewall's rule set/logs for denied packets destined to the app server's IP and port.
2. Confirm the rule order — an earlier "deny all" rule may be catching traffic before a later "allow" rule is reached.
3. Verify the correct source/destination zones, IPs, ports, and protocol (TCP/UDP) are specified in the allow rule.
4. Test connectivity with telnet/Test-NetConnection/nc to the specific port to confirm whether it's a full block or protocol/port-specific.
5. Check for NAT or routing misconfiguration introduced alongside the firewall (e.g., traffic reaching the firewall but not being forwarded).
6. Temporarily create an explicit allow rule for the app server's traffic, test, then tighten it to least-privilege.
7. Document and retest after each change until access is restored while internet access remains functional.

Section B- Practical Application

Task 1

Workstation Assembly

Prep and safety — Work on an anti-static mat or wear an anti-static wrist strap grounded to the case. Have the case, PSU, and all components laid out and unboxed.

Install the CPU — Open the CPU socket lever on the motherboard, align the CPU's notch/triangle marker with the

socket marker, drop it in gently (no force needed), and close the retention lever.

Apply thermal paste and install the cooler — Apply a small pea-sized dot of thermal paste to the CPU. Mount the cooler/heatsink, securing it evenly (cross-pattern if using screws) so pressure is even across the CPU. Connect the CPU fan header to the motherboard (usually labeled CPU_FAN).

Install RAM — Open the RAM slot clips, align the notch on the stick with the slot, and press down firmly on both ends until the clips snap shut. Use the motherboard manual to pick the correct slots for dual-channel (usually slots 2 and 4, not 1 and 2).

Mount the motherboard in the case — Install the I/O shield first, then line up the motherboard's screw holes with the case standoffs, and secure with screws (don't overtighten).

Install storage — Mount the SSD/HDD in its drive bay or M.2 slot. For M.2, insert at an angle and screw down flat. For 2.5"/3.5" drives, screw into the bay.

Install the PSU — Mount the power supply in its bay (fan facing the correct direction for your case's airflow) and screw it in.

Connect power cables

- 24-pin ATX cable → motherboard
- 4/8-pin CPU power cable → motherboard (near the CPU)
- SATA or PCIe power → storage drive and GPU (if used)
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Connect data cables — SATA data cable from the motherboard to the storage drive (if not M.2/NVMe).

Connect front panel headers — Match power switch, reset switch, power LED, HDD LED, and front-panel USB/audio connectors to the motherboard pins per the manual (this is the fiddliest step — go one pin group at a time).

Install any expansion cards (GPU, network card, etc.) into the appropriate PCIe slot and secure with a bracket screw; connect PCIe power if required.

Cable management — Route and tie down excess cables for airflow (not required for grading, but good practice to document).

Power on and verify POST — Connect a monitor and keyboard, power on, and confirm you see the manufacturer splash screen / BIOS POST beep or message. No POST errors = task complete.

Task 2

RAM Mismatch Troubleshooting

Power on and observe POST — Boot the system and watch for a RAM-related error message or beep code (e.g., repeated long beeps, or an on-screen message like "Memory Mismatch Error" or "Memory Size Decreased").

Enter BIOS/UEFI — Press the setup key during boot (Del, F2, or F10 depending on the motherboard) to access BIOS/UEFI.

Check the memory info screen — Go to the BIOS/UEFI's memory or hardware monitor section. Note each installed stick's detected speed, timing (CL/latency), and capacity.

Identify the mismatch — Compare the sticks against each other and against the motherboard's supported specs (from the manual/QVL list). Common mismatches:

- Different rated speeds (e.g., one stick DDR4-3200, another DDR4-2666) — BIOS will usually run all sticks at the slowest one's speed, or fail to boot if too incompatible.
- Different capacities in a dual-channel pair (not always fatal, but can disable dual-channel mode).
- Different CAS latency/timings, which can cause instability.

Determine root cause — Confirm whether the issue is due to mixing different brand/spec RAM, incorrect slot placement (not

using matched dual-channel slots), or a stick not seated properly (a common false "mismatch" cause).

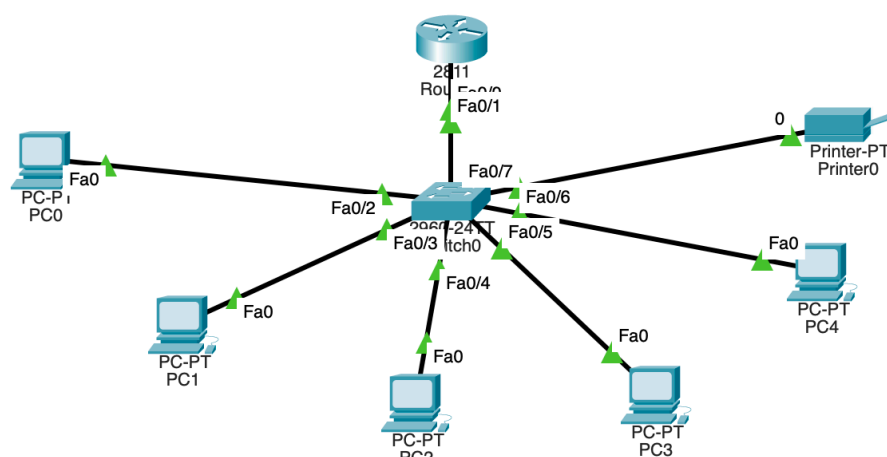
Reseat the RAM — Power off, unplug, discharge static, remove each stick, and firmly reseat it in the correct slot (check motherboard manual for the correct dual-channel slot pairing, e.g., A2/B2).

Reconfigure in BIOS if needed — If sticks are simply different speeds, manually set the memory frequency/timings in BIOS to the slower stick's rated speed (or enable XMP/DOCP if using identical matched kits) so both sticks run stable at a common value.

Save and reboot — Save BIOS settings, exit, and let the system POST again.

Confirm resolution — Check the BIOS memory screen again (or use dxdiag / lscpu+free -h / Task Manager once in the OS) to confirm the RAM is now detected correctly, at matching speed/timings, and the system boots without error messages.

Task 3



Section C - Mini Capstone: Diagnosing Random Reboots

Three likely hardware/thermal culprits:

1. **CPU** — thermal throttling or shutdown protection triggering under sustained load in a poorly ventilated cabinet.
2. **PSU** — under heavy load (GPU/CPU draw during rendering), a marginal or aging PSU can't sustain output, causing voltage sag and reboot.
3. **RAM** — heat-induced instability (less common than CPU/PSU, but rendering workloads stress memory too, and RAM errors under thermal stress can trigger crashes/reboots).

Why they pass POST in the morning but fail in the afternoon:

POST is a near-idle, momentary test — it doesn't load the CPU/GPU/PSU enough to reveal a thermal or power problem. Overnight, the cabinet and components cool to ambient temperature, so morning startup succeeds. As rendering workloads run through the afternoon, heat builds up inside the poorly ventilated cabinet (heat has nowhere to escape), component temperatures climb, and thermal protection (or PSU strain under sustained high draw) triggers a shutdown/reboot — a fault that only appears once heat and load accumulate.

Using a stress test to confirm the cause:

Run a combined CPU+GPU stress test while monitoring temperatures and voltages with a tool like HWInfo or Core Temp. If the reboot reliably reproduces once core temps approach thermal limits, it points to a thermal issue; if temps stay safe but the system reboots when power draw peaks, it points to the PSU being unable to sustain load (voltage rails sagging below spec).

Three remediation steps:

1. Improve airflow — relocate the workstations out of the closed cabinets, or add active cabinet ventilation/exhaust fans, and ensure case fans/heatsinks are clean and properly seated.
2. Reapply thermal paste and verify CPU cooler mounting pressure/contact.

3. Monitor and, if needed, undervolt or set more conservative fan curves/power limits as a stopgap while the ventilation fix is implemented.

PSU upgrade — necessary or not?

A PSU upgrade is likely **not** the first fix needed if temperatures are clearly hitting thermal limits during stress testing — the root cause is heat, not raw power capacity, and better ventilation should resolve it. However, if stress testing shows voltages sagging or the PSU is old/underrated for the combined CPU+GPU draw during rendering (i.e., it's running near or over its rated wattage), an upgrade **would** be justified, since a marginal PSU degrades further under heat and won't reliably support sustained rendering loads regardless of airflow improvements.

Section D - AI Augmented Learning

Step 1 — Prompt given to AI tool

"Generate an IP addressing and subnetting plan for a small office network with 5 systems and 1 network printer, using a Class C subnet. Specify the subnet, subnet mask, usable host range, and broadcast address. Also identify which addresses should be reserved for future expansion."

AI's Output

- Network: 192.168.1.0
- Subnet mask: 255.255.255.0 (/24)
- Usable host range: 192.168.1.1 – 192.168.1.254
- Broadcast address: 192.168.1.255
- Assigned: PC1–PC5 → .1–.5, Printer → .6
- "Reserved for future expansion": .7–.254

Step 2 — Corrections I'd make working without AI:

A full /24 for 6 devices is massively oversized and wastes address space — a proper subnetting exercise should right-size the subnet:

- Corrected subnet: **192.168.1.0/29** (255.255.255.248)

- Usable host range: 192.168.1.1 – 192.168.1.6 (6 usable hosts — exactly fits 5 PCs + 1 printer)
- Network address: 192.168.1.0, Broadcast address: 192.168.1.7
- Assignment: PC1–PC5 → .1–.5, Printer → .6
- For future expansion, reserve the **next block**, 192.168.1.8/29 (.9–.14), rather than leaving 248 addresses unused in the same broadcast domain.

What I changed and why:

"The AI's plan used a full /24 (254 usable addresses) for only 6 devices, which is inefficient and doesn't reflect proper subnetting practice — it also mislabeled 'reserved for expansion' as leftover addresses within the same oversized subnet rather than a deliberately allocated block. I corrected the plan to a /29 subnet, which exactly fits the 6 required hosts, and designated the next /29 block as the reserved range for future growth. This avoids wasting address space while still leaving room to expand."